LEARNING MODULE

20

Estimation Theory and Maximum Likelihood

An estimate is one draw from a sampling distribution. Its standard error says how far another sample could move it: TAIEX's 23.8 years pin the volatility to 18.3%–19.6% but leave the mean anywhere from 7.8% to 23.0% a year (95% intervals).

VOLUME IV · Statistical Inference and Econometrics

BUILDS ON LM1 · LM2 · LM4 · LM5 · LM7 · LM10 · LM11 · LM12 · LM13 · LM14 · LM15 · LM16 · LM17 · LM18 · LM19

LEADS TO LM21 · LM22 · LM23 · LM24 · LM25 · LM26 · LM27 · LM28 · LM30 · LM32 · LM33 · LM45

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	1 explain an estimator as a random variable with a sampling distribution, and calculate its bias, variance and mean squared error, including the trade-off of a scaled mean
☐	2 derive the standard error $\sigma/\sqrt{T}$ of a sample mean and the bias of s^2, and explain why the standard error of an annualized mean depends on the years of data, not on the sampling frequency
☐	3 construct and interpret confidence intervals for a mean, with the normal approximation and with the t_{T-1} interval of a normal model, and check their coverage by simulation
☐	4 derive the maximum-likelihood estimators of the normal model, and fit a Student-t by maximum likelihood numerically
☐	5 compute standard errors from the observed information, and state asymptotic normality and the Cramér–Rao bound with their conditions
☐	6 apply the delta method to the annualized volatility, the Kelly fraction and the Sharpe ratio of independent returns
☐	7 estimate a tail index with the Hill estimator and its standard error, and explain how the threshold trades bias against variance
☐	8 compute estimates, standard errors and intervals in Python with explicit divisors and seeded simulations

LEARNING MODULE OVERVIEW

- An estimator is a random variable, and its mean squared error is its variance plus its squared bias, (20.1). A biased estimator can win: with one year of TAIEX-like days, $0.397\bar{R}$ has 0.397 times the mean squared error of $\bar{R}$.
- The standard error of a mean is $\sigma/\sqrt{T}$, (20.2). Annualized it is $\sqrt{A}\,\sigma/\sqrt{\tau}$: the years of data decide it, not the sampling frequency. TAIEX's 23.8 years give 3.89 points; ±1 point at 95% would need 1,381 years.
- The interval $\bar{R} \pm 1.96\, s/\sqrt{T}$ is a large-sample one; for normal days the t_{T-1} interval with s_{u} is exact, (20.4). At $T = 5$ the first covers the mean in 84.5% of simulated samples, the second in 95.0%.
- Maximum likelihood gives $\bar{R}$ and s (divisor T) for the normal model, (20.5). For TAIEX it fits a Student-t with $\nu = 3.065$ (standard error 0.144), whose log-likelihood exceeds the normal's by 561.8.
- The observed information gives standard errors, (20.6), and no unbiased estimator beats the Cramér–Rao bound, (20.7). The profile log-likelihood puts ν between 2.80 and 3.37.
- The delta method, (20.8) and (20.9), gives TAIEX's annual volatility 18.96% (SE 0.33), the Kelly fraction 4.28 (SE 1.11; 2.10 to 6.46) and the Sharpe ratio 0.81 (SE 0.21) a year, all under independent days and a finite fourth moment.
- The Hill estimator, (20.10), puts TAIEX's tail index at 3.63 (SE 0.35) from the 108 losses of 3% or more. It rises from 2.49 at 2% to 4.93 at 4%: the threshold trades bias against variance.

1 INTRODUCTION

Every number computed from data is an estimate: another sample would give another value. This module measures how far estimates move between samples, and how to choose an estimator.

MOTIVATING CASE **What 23.8 years pin down**

From 3 January 2003 to 8 October 2026 the TAIEX total-return index, dividends reinvested, had calendar years from −43.07% (2008) to +83.34% (2009) (LM14). Its average day, 0.0625%, is 15.38% a year at $A = 246.12$ sessions. Its daily standard deviation, 1.2084%, is 18.96% a year (LM19).

Treat the 5,849 days as independent draws from one distribution. They then pin the volatility to 18.3% to 19.6% a year, but the mean only to 7.8% to 23.0%, both at 95% (Exhibit 1). Five-minute data would not narrow that range; only more years would. Why is the mean so much harder to estimate? And how far can the Kelly fraction of 4.28 that these numbers suggest be trusted?

Exhibit 1: Six TAIEX estimates and their standard errors

Quantity	Estimate	SE	95% interval	SE from
Mean, % a year, $A\bar{R}$	15.38	3.89	7.8 to 23.0	(20.2)
Volatility, % a year, $\sqrt{A}\,s$	18.96	0.33	18.3 to 19.6	delta method, kurtosis
Kelly fraction $\hat{f} = \bar{R}/s^2$	4.28	1.11	2.10 to 6.46	delta method, moments
Sharpe ratio a year, $r_f = 0$	0.81	0.21	0.40 to 1.22	delta method
Student-t degrees of freedom	3.065	0.144	2.78 to 3.35	observed information
Tail index, Hill, 108 losses of 3% or more	3.63	0.35	2.9 to 4.3	Pareto information

Note: TR23's daily simple returns, in-sample. Each standard error assumes iid days and states its model; an interval is the estimate ± 1.96 standard errors. The volatility's, the Kelly fraction's and the Sharpe ratio's standard errors also need a finite fourth moment, which the Student-t fit lacks (Section 7); they are nominal. The first, second and fourth estimates are LM19's. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026. Computations: `data/lm20_numbers.py` and `LM20_lab.ipynb`.

Sections 2 to 4 treat estimators and intervals, 5 and 6 maximum likelihood, 7 the delta method, 8 the Hill estimator and 9 Python.

> **NOTE** **Notation for this module**
>
> The data are TR23, the TAIEX total-return index: $T = 5{,}849$ daily simple returns R_t (a count) and log returns $r_t = \ln(1 + R_t)$. They span 8,680 calendar days from the first close, so $\tau = 8{,}680/365.25 = 23.7645$ years and $A = T/\tau = 246.12$ sessions a year. The moments $\bar{R}$, s, $\bar{r}$, s_r, Skew and Kurt divide by T, and s_{u} by $T - 1$.
>
> Local: $f_R(x;\theta)$ a model's density of R; H the Hessian of ℓ; $\hat{f} := \bar{R}/s^2$, the plug-in Kelly fraction; $\widehat{\mathrm{SR}} := \bar{R}/s$ ($r_f = 0$); η, η^* a constant multiplying $\bar{R}$ and its best value; $X = -R$ a day's loss (Section 8); (X, Y) a standard normal pair (Problem 6); $\bar{X}$ the mean of experiment 3's five draws (Problem 11); n the draws in LM18's $s_Y/\sqrt{2n}$; m days in a sum, $T_m = \lfloor T/m \rfloor$ such sums, s_m their standard deviation (divisor T_m); $Y \sim t_{T-1}$ in (20.4), and $V \sim \chi^2_{T-1}$ in its proof; $h(\boldsymbol{\theta})$ a smooth function; x_0 a loss level (the threshold), k the number of losses $x_i = -R_t$ beyond it; λ a Poisson mean and j a count of days (Problems 8, 9); L_t and (16.8)'s ε in Problem 6. Here α is a tail index (LM15), so a confidence level is a percentage.

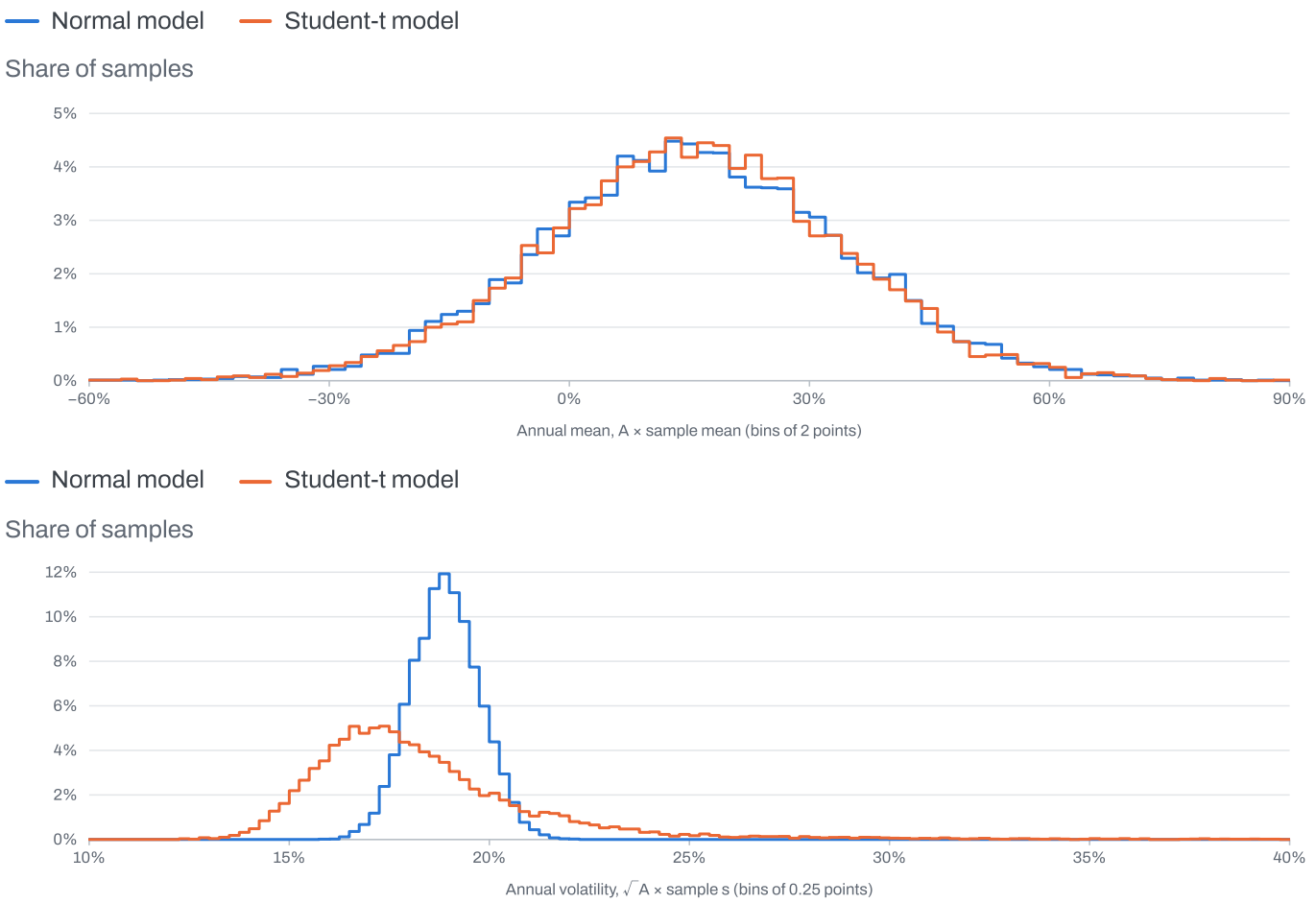
> **NOTE** **Research hygiene**
>
> Every TAIEX figure describes the total-return index, 3 January 2003 to 8 October 2026 (5,849 daily returns), except the 23 calendar years 2003–2025 (2003 from its first close). Every estimate, standard error and interval is **in-sample** and assumes independent, identically distributed days unless it says otherwise; none is a forecast. Simulated samples are **models**. The two price-limit regimes are a **stability check**. This module runs no tests (LM21).

2 ESTIMATORS, BIAS AND MEAN SQUARED ERROR

- ☐ explain an estimator as a random variable with a sampling distribution, and calculate its bias, variance and mean squared error, including the trade-off of a scaled mean

Before the data are drawn, the sample mean is a random variable: each possible sample gives its own value. Exhibit 2 draws 10,000 one-year samples of 246 days from two models with TR23's mean and standard deviation, a normal and LM15's Student-t, and estimates each sample's annual mean and volatility. Whether TAIEX's days are independent is a question about data, which LM19 answers; here the model assumes it.

Exhibit 2: One year of data, 10,000 times: the spread of two estimates



Note: experiment 1, `default_rng([1, 20261011])`: 10,000 samples of 246 days from a normal model, then 10,000 from a Student-t, both with $\bar{R} = 0.0625\%$ and $s = 1.2084\%$. The Student-t days are NumPy's `standard_t` with $\nu = 3.065257$, Exhibit 6's unrounded estimate, scaled by (15.11). Top: A times each sample's mean, bins of 2 points; 4 Student-t annual means outside −60% to 90% lie off the top axis.

Bottom: $\sqrt{A}$ times each sample's s (divisor 246), bins of 0.25 points; 35 Student-t values above 40% lie off the axis. A model, not data.

Monte Carlo standard errors (LM18's $s_Y/\sqrt{2n}$): 0.13 for the normal model's 18.9 and 0.006 for its 0.86. The Student-t has no finite fourth moment, so its two standard deviations get a range instead: over 20 further runs, 18.6 to 19.2 and 3.3 to 4.6. Experiment 4, `default_rng([4, 20261011])`, draws these runs in turn. LM18's batch standard errors (20 batches of 500) of the quantiles are 0.01 to 0.03 points, and 0.3 for the Student-t's 97.5% point.

The annual mean's standard deviation depends only on the variance: 18.9 points for the normal model, 18.7 for the Student-t and 18.96 in theory. The volatility's depends on the tails: 0.86 points for normal days (theory 0.855), about 4 for Student-t days. Its 95% range, $\hat{F}^{-1}(0.025)$ to $\hat{F}^{-1}(0.975)$, is 17.3% to 20.6% (normal) and 14.6% to 26.8% (Student-t, median 17.7%).

DEFINITION **Estimator, bias and mean squared error**

An **estimator** $\hat{\theta}$ of a parameter θ is a function of the sample, and its value on the data is the **estimate**. Its distribution across samples is its **sampling distribution**, and its standard deviation the **standard error** $\mathrm{SE}(\hat{\theta})$. The **bias** is $\mathrm{Bias}(\hat{\theta}) := \mathbb{E}\hat{\theta} - \theta$; the estimator is **unbiased** when the bias is 0. The **mean squared error** is $\mathrm{MSE}(\hat{\theta}) := \mathbb{E}[(\hat{\theta} - \theta)^2]$.

RESULT **Mean squared error**

If $\mathbb{E}\hat{\theta}^2 < \infty$, then

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2. \tag{20.1}$$

Proof. Write $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}\hat{\theta}) + \mathrm{Bias}(\hat{\theta})$ and square. The cross term is the constant bias times $\hat{\theta} - \mathbb{E}\hat{\theta}$, whose expectation is 0 by (14.3). ■

EXAMPLE 1

Shrinking the sample mean

Let T iid days have mean μ and standard deviation σ. Find the MSE of $\eta\bar{R}$ and the η that minimizes it. Evaluate η^* for one year ($T = 246$) and for TR23's $T = 5{,}849$, with TR23's daily $\bar{R}/s = 0.051727$ as the model's $\mathrm{SR} = \mu/\sigma$.

SOLUTION

By (14.3), (14.4) and (17.1), $\mathbb{E}(\eta\bar{R}) = \eta\mu$ and $\mathrm{Var}(\eta\bar{R}) = \eta^2\sigma^2/T$, so Bias $= (\eta - 1)\mu$. By (20.1),

$$\mathrm{MSE}(\eta\bar{R}) = \frac{\eta^2\sigma^2}{T} + (\eta - 1)^2\mu^2.$$

The derivative in η vanishes at

$$\eta^* = \frac{\mu^2}{\mu^2 + \sigma^2/T} = \frac{T\,\mathrm{SR}^2}{1 + T\,\mathrm{SR}^2},$$
$$\mathrm{MSE}(\eta^*\bar{R}) = \eta^*\,\frac{\sigma^2}{T}.$$

For one year, $T\,\mathrm{SR}^2 = 246 \times 0.051727^2 = 0.658$ and $\eta^* = 0.397$: the shrunk mean has 0.397 times the mean squared error of $\bar{R}$. For the whole sample, $T\,\mathrm{SR}^2 = 15.65$ and $\eta^* = 0.940$. The best η depends on the unknown μ; LM24 supplies it from a prior.

An estimator is **consistent** when it converges in probability to θ as $T \to \infty$ (LM17). By (17.2), $\bar{R}$ is consistent for iid days with finite variance. The strong law, applied to R_t and to R_t^2, makes $s^2 =$

$\frac{1}{T}\sum_t R_t^2 - \bar{R}^2$ converge to σ^2, so s is consistent too.

PITFALL **Unbiased is not the same as best**

An unbiased estimator is right on average but can still spread widely: in Example 1 the biased $\eta^*\bar{R}$ beats $\bar{R}$. Report the bias and the standard error separately.

導讀 **估計量是隨機變數**

樣本抽出之前，樣本平均數（sample mean）本身就是隨機變數（random variable）；用樣本算參數（parameter）的公式叫估計量（estimator），它在不同樣本間的分布（distribution）叫抽樣分布（sampling distribution）。估計量的期望值（expectation）減去真值是偏誤（bias），偏誤為零的叫不偏估計量（unbiased estimator）。好壞常用均方誤差（mean squared error）衡量，它等於變異數（variance）加偏誤的平方，所以稍有偏誤、變異數卻小得多的估計量可能更好：一年的資料下，$0.397\bar{R}$ 的均方誤差只有 $\bar{R}$ 的 0.397 倍。樣本越大越接近真值的叫一致估計量（consistent estimator）。模擬一萬個一年的樣本，年化平均報酬（annualized mean return）的標準差（standard deviation）約 19 個百分點，與尾部（tails）無關；年化波動率（annualized volatility）的分散卻由尾部決定。

KNOWLEDGE CHECK 1

For iid days, show that $\frac{1}{2}\bar{R}$ has a smaller MSE than $\bar{R}$ exactly when $|\mu/\sigma| < \sqrt{3/T}$. Does it, for one year of days with TR23's daily $\bar{R}/s = 0.0517$ as μ/σ?

ANSWER

By Example 1 with $\eta = \frac{1}{2}$, the condition is $\sigma^2/(4T) + \mu^2/4 < \sigma^2/T$, that is $\mu^2/\sigma^2 < 3/T$. For $T = 246$, $\sqrt{3/246} = 0.110$, above 0.0517, so yes, if the model's μ and σ were TR23's.

3 THE STANDARD ERROR OF A MEAN

- ☐ derive the standard error $\sigma/\sqrt{T}$ of a sample mean and the bias of s^2, and explain why the standard error of an annualized mean depends on the years of data, not on the sampling frequency

For uncorrelated days with a common mean and variance, the sum rule (16.3) and (17.1) give $\operatorname{Var}\bar{R} = \sigma^2/T$. A mean over $\tau = T/A$ years scales by A, so its standard error does too.

RESULT **Standard error of a mean**

Let $R_1, \dots, R_T$ be uncorrelated with a common mean μ and a common variance $\sigma^2 < \infty$. Then

$$\begin{aligned} \mathrm{SE}(\bar{R}) &= \frac{\sigma}{\sqrt{T}}, \\ \mathrm{SE}(A\bar{R}) &= \frac{A\sigma}{\sqrt{T}} = \frac{\sqrt{A}\,\sigma}{\sqrt{\tau}}, \end{aligned} \tag{20.2}$$

estimated by replacing σ with s.

EXAMPLE 2

The standard error of TAIEX's mean

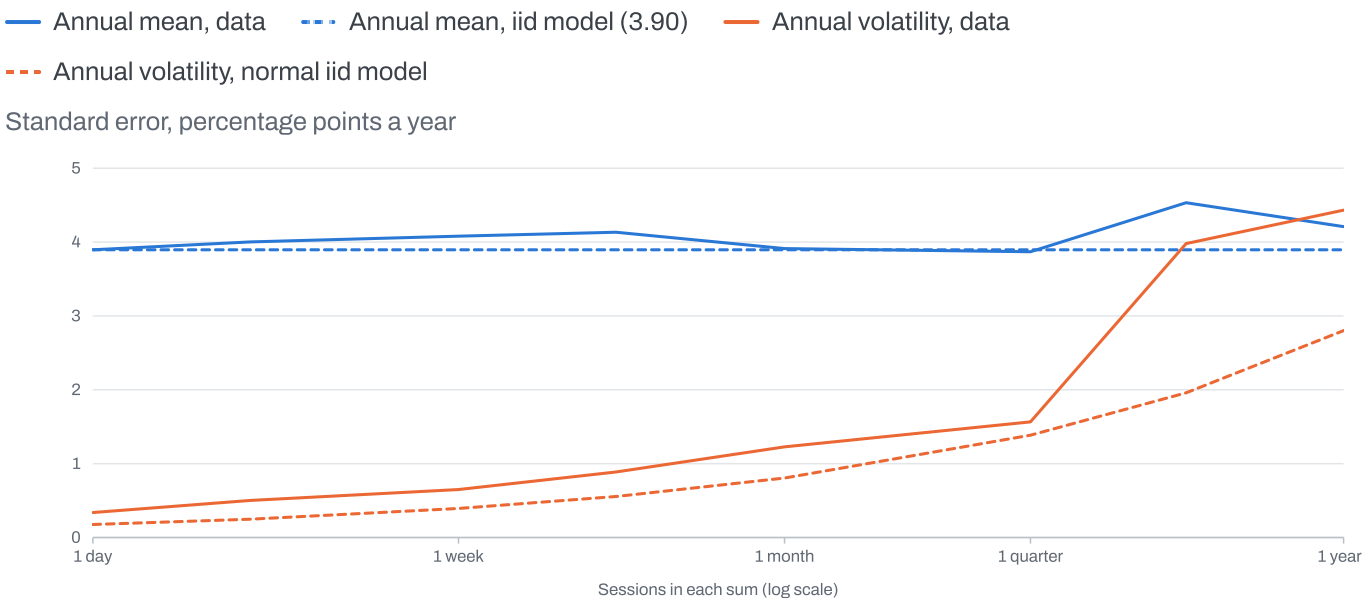
Compute the standard error of TR23's daily mean and of its annual mean $A\bar{R} = 15.38\%$.

SOLUTION

By (20.2), $s/\sqrt{T} = 1.2084\%/\sqrt{5{,}849} = 0.01580\%$ a day, the value that LM18's (18.8) gave for its bootstrap model. Times $A = 246.12$, it is 3.89 points a year, a quarter of the 15.38%. Equivalently, $18.96\%/\sqrt{23.7645} = 3.89$. LM7's 7.36 and 6.09 points over 4.95 years follow the same rule.

Merton (1980) drew the consequence: finer sampling does not help the mean. By (1.12), $\sum_t r_t = \ln(P_T/P_0)$, so the annual mean log return $A\bar{r} = \ln(P_T/P_0)/\tau$ depends only on the first and last price. Five-minute log returns over the same years give the same number.

Exhibit 3: Standard errors of the annual mean and volatility, by sampling interval



Note: TR23's log returns summed over non-overlapping blocks of m sessions from the first return, T_m sums (the last $T - mT_m$ returns left out). Mean: $(A/m)\, s_m/\sqrt{T_m}$, against the iid line $\sqrt{A}\, s_r/\sqrt{\tau} = 3.90$. Volatility: Section 7's delta method with the sums' own kurtosis, against the normal iid line $\sqrt{A}\, s_r/\sqrt{2T_m}$. In-sample; each point rests on one partition. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026.

Exhibit 3 measures both standard errors at eight sampling intervals, from one day to one year. The mean's stays between 3.87 and 4.53 points, around the iid line. The volatility's grows from 0.34 points with daily data to 4.43 with annual sums, and the normal model's from 0.18 to 2.80. Fine sampling sharpens the volatility, never the mean. A mean known to ±1 point at 95% needs $(1.96 \times 18.96)^2 = 1{,}381$ years of data.

The mean's standard error survives volatility clustering if the days are uncorrelated: with unequal variances σ_t^2, (16.3) gives $\operatorname{Var} \bar{R} = \frac{1}{T^2} \sum_t \sigma_t^2$. Correlated days break (20.2). LM19's variance ratios measure how far daily variances fail to add up; LM23's HAC errors and LM22's block bootstrap then repair the standard error.

An expansion around μ gives the bias of s^2. Since $\sum_t (R_t - \bar{R}) = 0$,

$$\sum_{t=1}^{T} (R_t - \mu)^2 = \sum_{t=1}^{T} (R_t - \bar{R})^2 + T(\bar{R} - \mu)^2.$$

The left side has expectation $T\sigma^2$, and by (20.2) the last term has σ^2, so $\mathbb{E} \sum_t (R_t - \bar{R})^2 = (T-1)\sigma^2$.

RESULT **The bias of the sample variance**

Under the conditions of (20.2),

$$\mathbb{E}s^2 = \frac{T-1}{T}\,\sigma^2, \qquad s_{\mathrm{u}}^2 := \frac{T}{T-1}\,s^2, \qquad \mathbb{E}s_{\mathrm{u}}^2 = \sigma^2. \tag{20.3}$$

The conditions of (20.2) are those under which LM10's divisor $T-1$ removes a small bias. NumPy's `ddof=1` and Polars' default compute s_{u}, which LM19 introduced.

EXAMPLE 3

How much the divisor matters

Compute s_{u}/s for TR23's 5,849 days and for the 23 annual log returns of 2003–2025, whose s is 22.207% (LM15).

SOLUTION

The ratio is $\sqrt{T/(T-1)}$: 1.0000855 for TR23, a change in the fifth significant figure. For 23 values it is $\sqrt{23/22} = 1.0225$, so $s_{\mathrm{u}} = 22.706\%$, and $\mathbb{E}s^2$ is only $22/23 = 0.9565$ of σ^2. LM10's 4,382 returns gave 1.000114. The divisor matters for calendar years and other small samples, not for daily data.

PITFALL **Annualizing the mean's error by the square root**

A mean scales by A, so its standard error does too. The daily 0.0158% times $\sqrt{A}$ gives 0.25 points, $\sqrt{A} = 15.69$ times less than the correct 3.89. Only a standard deviation scales by $\sqrt{A}$, and exactly only for log returns on uncorrelated days (LM19).

導讀 **平均數的標準誤只看年數**

樣本平均數（sample mean）的標準誤（standard error）是 $\sigma/\sqrt{T}$，年化之後等於年化波動率（annualized volatility）除以資料年數的平方根，與抽樣頻率（sampling frequency）無關。加權股價報酬指數 23.8 年的資料，年化平均報酬（annualized mean return）的標準誤約 3.89 個百分點；改用五分鐘資料也不會變小，因為對數報酬（log return）的總和只取決於頭尾兩個價格。要讓年化平均報酬的 95% 信賴區間（confidence interval）縮到正負 1 個百分點，需要約 1,381 年的資料。樣本變異數（sample variance）除以 T 時，期望值（expectation）只有 σ^2 的 $(T-1)/T$ 倍，除以 $T-1$ 才不偏（unbiased）；日資料只差在第五位有效數字（significant figure），23 個年報酬的標準差（standard deviation）則差 2.2%。

KNOWLEDGE CHECK 2

Three years of independent daily returns have an annual volatility of 18.96%. What is the standard error of the annual mean? Ten years? What if the returns were monthly over the same years?

ANSWER

By (20.2), $18.96/\sqrt{3} = 10.9$ points and $18.96/\sqrt{10} = 6.0$ points. Monthly data over the same years give the same, because only the years enter.

4 CONFIDENCE INTERVALS

- [ ] construct and interpret confidence intervals for a mean, with the normal approximation and with the t_{T-1} interval of a normal model, and check their coverage by simulation

A standard error becomes an interval once the shape of the estimator's distribution is known.

DEFINITION **Confidence interval**

A **95% confidence interval** for θ is an interval computed from the sample that contains θ with probability 0.95 for every θ, exactly or approximately for large T. The probability is over samples, with θ fixed; it is the interval's **coverage**.

By (17.5), $\sqrt{T}(\bar{R}-\mu)/\sigma$ is approximately standard normal for iid days with finite variance. Replacing σ by the consistent s keeps the approximation (stated without proof; Wasserman 2004). With $\Phi^{-1}(0.975)=1.96$, (15.6), this gives a large-sample interval; for normal days an exact one exists.

RESULT **Confidence intervals for a mean**

Let the days be iid with mean μ and variance $0<\sigma^2<\infty$. For large T, an approximate 95% interval for μ is $\bar{R}\pm 1.96\,s/\sqrt{T}$. If the days are iid normal and $T\geq 2$, an exact 95% interval is

$$\bar{R}\pm F_Y^{-1}(0.975)\,\frac{s_{\mathrm{u}}}{\sqrt{T}},\qquad Y\sim t_{T-1}. \tag{20.4}$$

Proof of the exact form. Let $Z=\sqrt{T}(\bar{R}-\mu)/\sigma$ and $V=(T-1)s_{\mathrm{u}}^2/\sigma^2$. For iid normal days, $Z\sim\mathcal{N}(0,1)$ and $V\sim\chi^2_{T-1}$ (LM16; Blitzstein and Hwang 2019, Example 10.4.3), and they are independent (their Example 7.5.9). By their Definition 10.4.4, whose density is (15.9) (they state it without proof),

$$\frac{Z}{\sqrt{V/(T-1)}}=\frac{\sqrt{T}(\bar{R}-\mu)}{s_{\mathrm{u}}}\sim t_{T-1},$$

so $|\bar{R}-\mu|\leq F_Y^{-1}(0.975)\,s_{\mathrm{u}}/\sqrt{T}$ with probability 0.95. ■

EXAMPLE 4

TAIEX's annual mean

Give 95% intervals for TR23's annual mean with the normal approximation and with the t_{T-1} form. How wide is the interval from one year of data with the same volatility?

SOLUTION

By (20.4) and Example 2, $15.38\%\pm 1.96\times 3.89$ runs from 7.8% to 23.0%. At $T-1=5{,}848$ the t quantile is 1.96037 against $\Phi^{-1}(0.975)=1.95996$, and $s_{\mathrm{u}}/s=1.0000855$, so the t interval differs by 0.002 points. One year of data gives a standard error of 18.96 points and a half-width of $1.96\times 18.96=37.2$ points.

Exhibit 4 checks both intervals by simulation. With s and 1.96, small samples cover too rarely, 84.5% at $T=5$, because the normal quantile ignores the error in s, and divisor T shrinks s. The t_{T-1} interval is exact for normal days and covers slightly too often for t_3 days. By $T=246$ every column is between 94.8% and 95.2%.

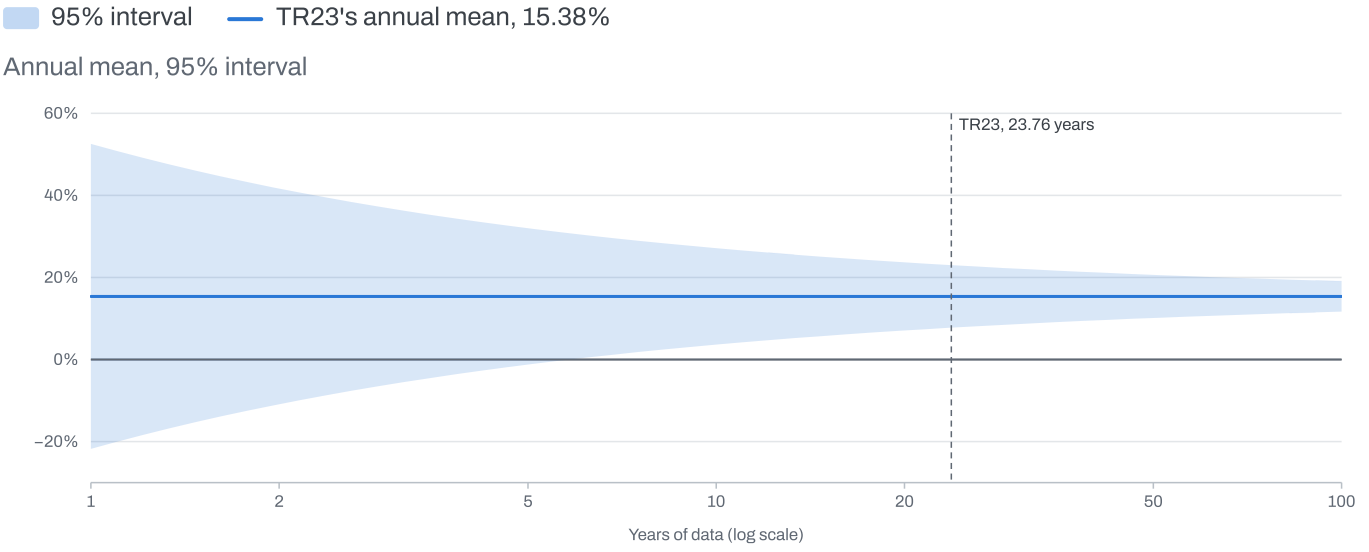
Exhibit 4: How often a 95% interval covers the mean

T	Normal days: 1.96 with s	exact	t_{T-1} with s_{u}	t_3 days: 1.96 with s	t_{T-1} with s_{u}
5	84.5	84.6	95.0	86.1	96.4
10	90.4	90.4	95.1	91.5	96.0
30	93.6	93.6	94.9	94.2	95.6
100	94.6	94.6	95.0	94.9	95.3
246	94.8	94.8	95.0	95.0	95.2

Note: coverage in %, experiment 2, `default_rng([2, 20261011])`: 100,000 samples of T iid days per cell, drawn row by row with the normal days first, in chunks of 10,000 (the lab's layout). Monte Carlo standard error 0.07 points at 95%. The exact column is $2F_{t_{T-1}}\left(1.96\sqrt{(T-1)/T}\right) - 1$, from the proof of (20.4), because $1.96\,s = 1.96\sqrt{(T-1)/T}\,s_{\mathrm{u}}$. The t_3 days are built from four standard normals and scaled to variance 1, as in the widget. A model, not data.

Exhibit 5 varies the years of data, with TR23's annual mean and volatility. The interval runs from −21.8% to 52.5% after one year, 3.6% to 27.1% after ten and 11.7% to 19.1% after a hundred.

Exhibit 5: The 95% interval for an annual mean against years of data



Note: $A\bar{R} \pm 1.96\sqrt{A}\,s/\sqrt{\text{years}}$ for 1 to 100 years of iid days, with TR23's $A\bar{R} = 15.38\%$ and $\sqrt{A}\,s = 18.96\%$ held fixed; log scale for years. At TR23's 23.76 years it is 7.8% to 23.0%. A model of hypothetical samples, not data.

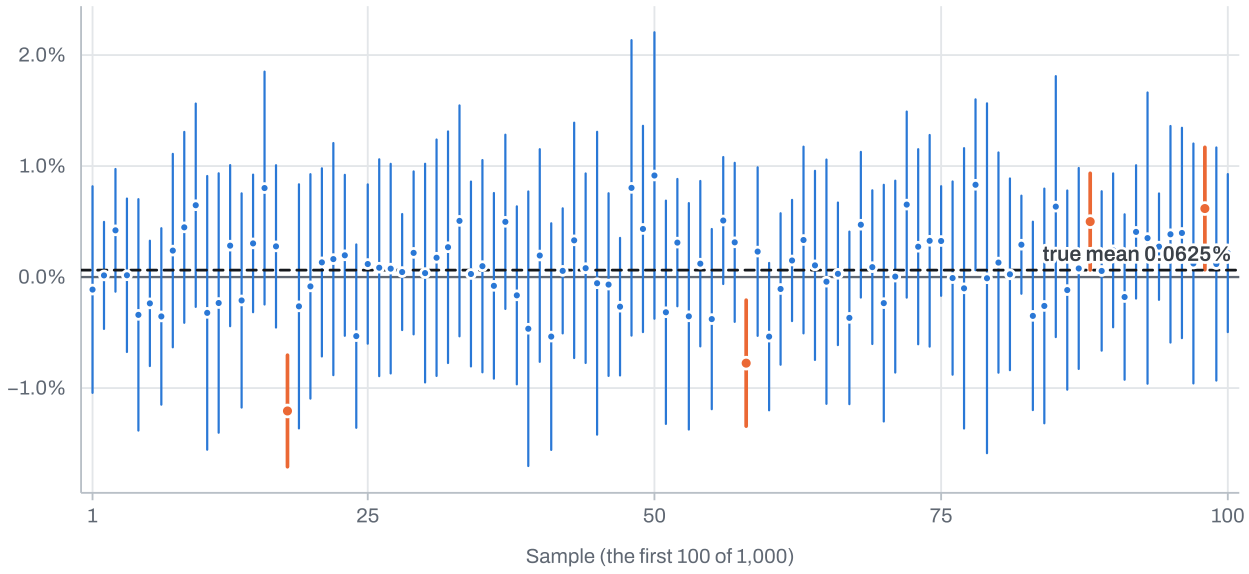
The widget repeats the experiment of Exhibit 4 with 1,000 samples per click and draws the first 100 intervals.

How often does a 95% interval cover the mean?

INTERACTIVE

Interval covers the mean Interval misses the mean

Mean daily return and its 95% interval



Coverage, 1.96 with s (divisor T)	Coverage, t quantile with s_u	t quantile, 9 degrees of freedom
91.4% ± 0.9	**95.4% ± 0.7**	**2.2622**
Exact coverage of 1.96 with s	**Samples · seed**	
90.4%	**1,000 · 1**	

A model, not data: each draw makes 1,000 samples of T independent days with TAIEX's mean, 0.0625% a day, and standard deviation, 1.2084%, either normal or Student-t with 3 degrees of freedom (four normal draws per day, scaled to the same variance). The chart shows the first 100 intervals of the chosen kind around their sample means. The generator is the book's seeded one, not NumPy's, so its numbers differ from the module's table; the text quotes none of them.

PITFALL The 95% belongs to the procedure

Once computed, 7.8% to 23.0% either contains $A\mu$ or does not. The 95% says that intervals built this way cover $A\mu$ in 95% of samples, if the days are iid. A probability for $A\mu$ itself needs LM24. An interval that excludes a value is the start of a test, LM21's subject; this module runs none.

KNOWLEDGE CHECK 3

At $T = 5{,}849$, does the t_{T-1} quantile change TR23's interval for the annual mean at two decimals?

ANSWER

No. The interval is 7.76% to 23.01% either way (7.7620 to 23.0059 against 7.7599 to 23.0080); the quantile is 1.96037 against 1.95996.

5 MAXIMUM LIKELIHOOD

- [ ] derive the maximum-likelihood estimators of the normal model, and fit a Student-t by maximum likelihood numerically

LM15 fitted a Student-t to TAIEX with SciPy's `t.fit`, by maximum likelihood. Let the days be iid with a density $f_R(x;\boldsymbol{\theta})$, known up to the parameters $\boldsymbol{\theta}$.

DEFINITION **Likelihood**

The **likelihood** of iid observations $R_1,\dots,R_T$ is $\mathcal{L}(\boldsymbol{\theta}) := \prod_{t=1}^{T} f_R(R_t;\boldsymbol{\theta})$, and the **log-likelihood** is $\ell(\boldsymbol{\theta}) := \ln \mathcal{L}(\boldsymbol{\theta}) = \sum_t \ln f_R(R_t;\boldsymbol{\theta})$. The **maximum-likelihood estimator** (MLE) $\hat{\boldsymbol{\theta}}$ maximizes ℓ over the parameters the model allows.

The likelihood holds the data fixed and ranks parameter values by the density they give the sample. For the normal model, $f_R(x;\mu,\sigma) = \varphi\big((x-\mu)/\sigma\big)/\sigma$, and Section 3's expansion gives

$$\begin{aligned}\ell(\mu,\sigma) = &-\frac{T}{2}\ln(2\pi) - T\ln\sigma \\ &-\frac{Ts^2 + T(\bar{R}-\mu)^2}{2\sigma^2}.\end{aligned}$$

RESULT **Normal maximum likelihood**

For iid normal days with $T \geq 2$, not all equal, the MLE is

$$\hat{\mu} = \bar{R}, \qquad \hat{\sigma} = s. \tag{20.5}$$

For iid $\mathcal{N}_N(\boldsymbol{\mu},\Sigma)$ vectors $\mathbf{R}_t$, the MLE is $\bar{\mathbf{R}}$ and LM10's $\hat{\Sigma}$ of (10.2), whenever $\hat{\Sigma}$ is positive definite (Deep Dive 1).

Proof (one variable). For every σ, ℓ is largest at $\mu = \bar{R}$, where it is $-\frac{T}{2}\ln(2\pi) - T\ln\sigma - Ts^2/(2\sigma^2)$. Its derivative in σ, $-T/\sigma + Ts^2/\sigma^3$, is positive below s and negative above. ■

So the maximum-likelihood variance s^2 divides by T and is biased, by (20.3): maximum likelihood aims at large-sample precision, not at unbiasedness.

EXAMPLE 5

LM5's normal fit is maximum likelihood

LM5 fitted $\mathcal{N}(\mu,\sigma^2)$ to 0050's 2,902 daily total returns with $\hat{\mu} = 0.0871\%$ and $\hat{\sigma} = s = 1.2385\%$. Is that the MLE? What is TR23's?

SOLUTION

Yes, by (20.5): LM5's "population standard deviation" of the sample divides by T. The mean's standard error is 0.0230% (LM4). For TR23 the MLE is $\hat{\mu} = 0.0625\%$ and $\hat{\sigma} = 1.2084\%$, not $s_{\mathrm{u}} = 1.2085\%$.

DEEP DIVE **Deep Dive 1: maximum likelihood for $\mathcal{N}_N$** optional · click to expand

Here K is a square matrix, L a Cholesky factor, (12.8), and λ_i, $i = 1, \ldots, N$, the eigenvalues of M below. By the density in (16.7), the log-likelihood of iid $\mathcal{N}_N(\boldsymbol{\mu}, \Sigma)$ vectors with Σ positive definite is

$$\begin{aligned}\ell = &-\frac{T}{2}\big(N \ln 2\pi + \ln\det\Sigma\big)\\ &-\frac{1}{2}\sum_{t=1}^{T}(\mathbf{R}_t - \boldsymbol{\mu})^{\top}\Sigma^{-1}(\mathbf{R}_t - \boldsymbol{\mu}).\end{aligned}$$

As in Section 3, the sum splits into $\sum_t (\mathbf{R}_t - \bar{\mathbf{R}})^{\top}\Sigma^{-1}(\mathbf{R}_t - \bar{\mathbf{R}})$ plus $T(\bar{\mathbf{R}} - \boldsymbol{\mu})^{\top}\Sigma^{-1}(\bar{\mathbf{R}} - \boldsymbol{\mu})$, since the cross terms sum to zero. The second part is positive unless $\boldsymbol{\mu} = \bar{\mathbf{R}}$, so $\hat{\boldsymbol{\mu}} = \bar{\mathbf{R}}$.

With $\mathbf{x}^{\top}K\mathbf{x} = \operatorname{tr}(K\mathbf{x}\mathbf{x}^{\top})$ and (10.2), the first part equals $T\operatorname{tr}(\Sigma^{-1}\hat{\Sigma})$. Let $\hat{\Sigma} = LL^{\top}$ be positive definite, which needs $T > N$, and let $M = L^{\top}\Sigma^{-1}L$, symmetric with eigenvalues $\lambda_i > 0$. Then $\operatorname{tr}(\Sigma^{-1}\hat{\Sigma}) = \operatorname{tr} M = \sum_i \lambda_i$ and $\det\Sigma = \det\hat{\Sigma}/\prod_i \lambda_i$ (LM11), so, at $\boldsymbol{\mu} = \bar{\mathbf{R}}$,

$$\begin{aligned}\ell = -\frac{T}{2}\Big[&N\ln 2\pi + \ln\det\hat{\Sigma}\\ &+\sum_{i=1}^{N}(\lambda_i - \ln\lambda_i)\Big].\end{aligned}$$

By (2.1), $\ln x \le x - 1$, so each $\lambda_i - \ln\lambda_i \ge 1$, with equality only at $\lambda_i = 1$. The maximum needs $M = I$, that is $\Sigma = LL^{\top} = \hat{\Sigma}$, with divisor T. ■

LM15's Student-t is the density (15.9), shifted by a location and stretched by a scale. No formula gives its MLE, so a numerical search maximizes ℓ. LM26 fits GARCH models with the same machinery.

EXAMPLE 6

A Student-t for TAIEX by maximum likelihood

Fit the Student-t to TR23's daily returns by maximum likelihood, and compare its maximal ℓ with the normal model's.

SOLUTION

SciPy's `t.fit` returns $\nu = 3.06522$, and a tighter Nelder–Mead search from there reaches 3.06526, location 0.0997% and scale 0.7761% (Exhibit 6). The maximal ℓ exceeds the normal model's by 561.8. By (15.11), the fit's standard deviation is $0.7761\% \times \sqrt{3.065/1.065} = 1.317\%$, against $s = 1.2084\%$.

Exhibit 6: Normal and Student-t fits to TAIEX's daily returns

Model, returns	T	ν (SE)	Location, % (SE)	Scale, % (SE)	Standard deviation, % (SE)
Normal, TR23	5,849	—	0.0625 (0.0158)	—	1.2084 (0.0112)
Student-t, TR23	5,849	3.065 (0.144)	0.0997 (0.0124)	0.7761 (0.0138)	1.317 (0.045)
Student-t, 7% limit, to 29 May 2015	3,077	2.990 (0.195)	—	—	—
Student-t, 10% limit, from 1 June 2015	2,772	3.237 (0.223)	—	—	—

Note: TR23's daily simple returns in percent; maximum likelihood, in-sample. Standard errors come from the observed information (Section 6), a central-difference Hessian of $-\ell$ that agrees at two step sizes; the standard deviation's from (20.8). The regimes are a stability check. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026.

PITFALL **The Student-t's location is not the mean**

The fit centres a symmetric model on skewed data (skewness −0.34). Its location, 0.0997%, sits near LM14's median, 0.0879% (its standard error is LM32's), and more than half above the mean, 0.0625%. Use $\bar{R}$ for the drift.

KNOWLEDGE CHECK 4

Why must the Student-t's maximal log-likelihood be at least the normal's, and what does the gain of 561.8 not yet show?

ANSWER

The normal is the Student-t's limit as $\nu \to \infty$, so the Student-t's maximum cannot be lower. Whether 561.8 exceeds what a third parameter buys by chance needs a likelihood-ratio test, which this book does not cover (Section 6).

6 FISHER INFORMATION AND THE CRAMÉR–RAO BOUND

- [ ] compute standard errors from the observed information, and state asymptotic normality and the Cramér–Rao bound with their conditions

A sharp peak in ℓ pins the parameter down; a flat top leaves many values almost as likely.

DEFINITION **Score and Fisher information**

For one observation with density $f_R(x;\theta)$, the **score** is $\partial_\theta \ln f_R(R;\theta)$ and the **Fisher information** is

$$\mathcal{I}(\theta) := -\mathbb{E}\Big[\frac{\partial^2}{\partial\theta^2}\ln f_R(R;\theta)\Big].$$

The **observed information** of a sample is $-\ell''(\hat\theta)$, and for a parameter vector the matrix $-H$ at $\hat{\boldsymbol\theta}$.

A sample of T iid observations carries $T\mathcal{I}(\theta)$, which the observed information estimates. Deep Dive 2 shows that $\mathcal{I}(\theta)$ is also the variance of the score.

THEOREM **Asymptotic normality of the MLE**

Let the following hold:

- the observations are iid with density $f_R(x;\theta)$, the model is correct, and distinct θ give distinct densities;
- the true θ is an interior point of the parameter set, and the support of f_R does not depend on θ;
- the density is smooth in θ, with $0 < \mathcal{I}(\theta) < \infty$.

Then, for large T,

$$\begin{aligned}\hat\theta &\approx \mathcal{N}\Big(\theta, \frac{1}{T\mathcal{I}(\theta)}\Big),\\ \mathrm{SE}(\hat\theta) &\approx \big(-\ell''(\hat\theta)\big)^{-1/2},\end{aligned} \tag{20.6}$$

and for a vector $\mathrm{Cov}(\hat{\boldsymbol\theta}) \approx (-H)^{-1}$ at $\hat{\boldsymbol\theta}$. Stated without proof (Wasserman 2004).

EXAMPLE 7

The normal model's information

Compute $\mathcal{I}(\mu)$ and $\mathcal{I}(\sigma)$ for $\mathcal{N}(\mu,\sigma^2)$, and the standard errors of TR23's $\hat\mu$ and $\hat\sigma$.

SOLUTION

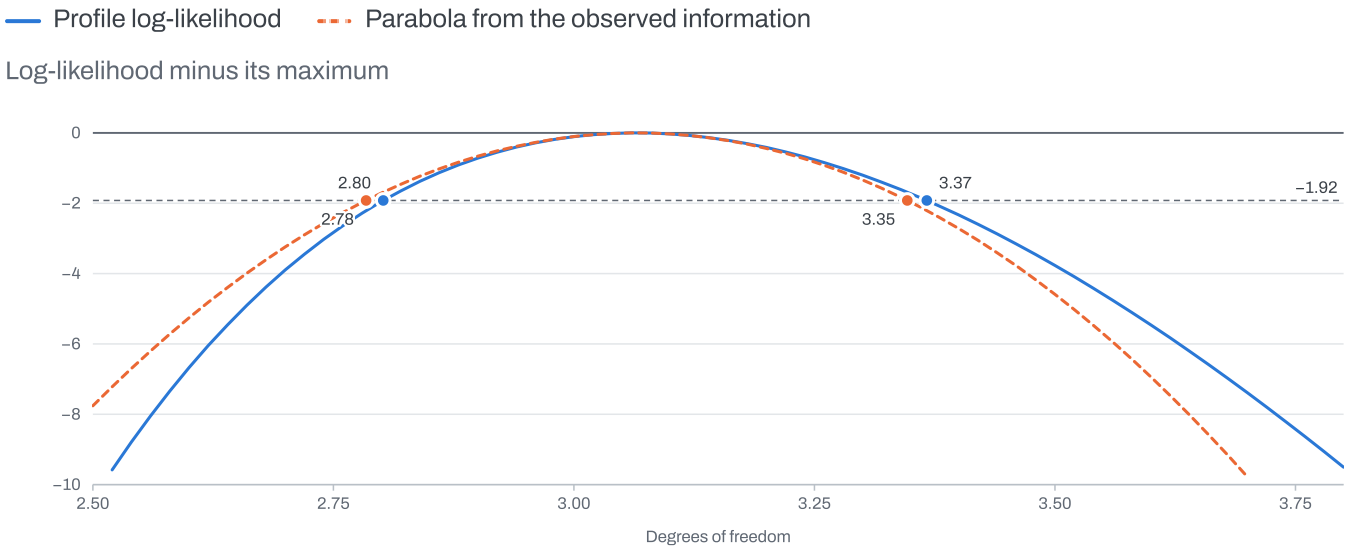
From $\ln f_R = -\ln\sigma - (x-\mu)^2/(2\sigma^2)$ plus a constant, $\partial^2_\mu \ln f_R = -1/\sigma^2$ and $\partial^2_\sigma \ln f_R = 1/\sigma^2 - 3(x-\mu)^2/\sigma^4$, whose expectation is $-2/\sigma^2$. The mixed derivative has expectation 0, so the two parameters can be treated one at a time: $\mathcal{I}(\mu) = 1/\sigma^2$ and $\mathcal{I}(\sigma) = 2/\sigma^2$. By (20.6),

$$\mathrm{SE}(\hat{\mu}) = \frac{s}{\sqrt{T}} = 0.0158\%,$$
$$\mathrm{SE}(\hat{\sigma}) = \frac{s}{\sqrt{2T}} = 0.0112\%.$$

The first is (20.2) again, and the second is LM18's $s_Y/\sqrt{2n}$ for near-normal draws.

The **profile log-likelihood** of one parameter maximizes ℓ over the others at each of its values. Exhibit 7 draws it for ν, beside the parabola of the observed information. Where the curve falls $1.96^2/2 = 1.92$ below its top it gives 2.80 to 3.37; the parabola gives $\hat{\nu} \pm 1.96 \times 0.144$, 2.78 to 3.35. Comparing two maximized log-likelihoods is a likelihood-ratio test, outside this book (Wasserman 2004, §10.6).

Exhibit 7: Profile log-likelihood of the Student-t's degrees of freedom



Note: TR23's daily returns; at each ν the location and scale are re-maximized. The dashed parabola has the curvature $1/0.144^2$ of the observed information. In-sample. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026.

The curve falls faster than the parabola towards small ν, so its interval is skewed. By $(-H)^{-1}$, the estimates of ν and of the scale correlate 0.68: fatter tails come with a smaller scale. The regimes' 2.990 (0.195) and 3.237 (0.223) in Exhibit 6 are a stability check.

THEOREM **Cramér–Rao bound**

Let the following hold:

- the conditions of (20.6);
- the identity $\int f_R(x;\theta)\,dx = 1$ may be differentiated twice under the integral sign, and $\mathbb{E}\hat{\theta}$ once;
- the estimator $\hat{\theta}$ of θ from T iid observations is unbiased.

Then

$$\operatorname{Var}(\hat{\theta}) \geq \frac{1}{T\mathcal{I}(\theta)}. \tag{20.7}$$

Deep Dive 2 proves the bound (Rao 1992, a reprint). In the normal model $\bar{R}$ attains it: $\operatorname{Var}\bar{R} = \sigma^2/T = 1/(T\mathcal{I}(\mu))$. Inside the Student-t model, $\bar{R}$'s standard error is the fit's $1.317\%/\sqrt{T} = 0.0172\%$, and the location's is 0.0124%, a ratio of 0.72. That efficiency holds only if the days are Student-t; on TR23's skewed days the location estimates a different centre from the mean (Section 5).

DEEP DIVE **Deep Dive 2: why no unbiased estimator beats $1/(T\mathcal{I})$** optional · click to expand

Here $\mathbf{x} = (x_1, \ldots, x_T)$ is a possible sample, $g(\mathbf{x};\theta) = \prod_t f_R(x_t;\theta)$ its joint density and $u = \partial_\theta \ln g$ the sample's score. Differentiating $\int g\,d\mathbf{x} = 1$ under the integral gives $\mathbb{E}u = \int \partial_\theta g\,d\mathbf{x} = 0$. Differentiating $\int (\partial_\theta \ln f_R)\, f_R\,dx = 0$ once more gives $\mathbb{E}[(\partial_\theta \ln f_R)^2] = \mathcal{I}(\theta)$, so the sum u of T iid scores has $\operatorname{Var} u = T\mathcal{I}(\theta)$ by (16.3).

Unbiasedness says $\int \hat{\theta}(\mathbf{x})\, g\,d\mathbf{x} = \theta$; differentiating gives $\mathbb{E}[\hat{\theta}u] = 1$, so $\operatorname{Cov}(\hat{\theta}, u) = 1$ since $\mathbb{E}u = 0$. For $\operatorname{Var}\hat{\theta} < \infty$, the bound $|\rho| \leq 1$ (LM16) gives $1 = \operatorname{Cov}(\hat{\theta}, u)^2 \leq \operatorname{Var}(\hat{\theta})\, T\mathcal{I}(\theta)$, which is (20.7). An infinite variance satisfies (20.7) trivially. ■

PITFALL **An information-based standard error assumes the model**

Formula (20.6) holds when the model is right. A wrong model can make $(-H)^{-1}$ far too small: Problem 8's Poisson standard error understates the spread of TAIEX's monthly counts of large moves.

導讀 **概度、訊息與下界**

把資料固定、讓參數（parameter）變動，樣本的聯合密度（joint density）就成了概度函數（likelihood function），使它最大的參數值就是最大概度估計（maximum likelihood estimation）。常態模型（normal model）的最大概度估計正是樣本平均數（sample mean）與除以 T 的標準差（standard deviation）；t 分布（Student-t distribution）則要用數值方法（numerical methods）求解，加權股價報酬指數的自由度（degrees of freedom）約為 3.065。單筆觀察的對數概度（log-likelihood）二階導數（second derivative）取負號後的期望值（expectation）是費雪訊息（Fisher information）；實務上用樣本對數概度在頂點的彎曲程度，即負的 Hessian 矩陣，也就是觀察訊息矩陣（observed information matrix），其反矩陣（inverse matrix）對角元素（diagonal entries）的平方根就是近似的標準誤（standard error）；樣本大時，估計量（estimator）具漸近常態性（asymptotic normality）。克萊姆-饒歐下界（Cramér–Rao lower bound）說不偏估計量（unbiased estimator）的變異數（variance）不會低於 $1/(T\mathcal{I})$，達到下界的估計量最有效率（efficiency）。剖面概度（profile likelihood）比拋物線（parabola）更忠實：自由度的區間（interval）2.80 到 3.37 並不對稱。

KNOWLEDGE CHECK 5

One year of iid normal days has the annual volatility of TR23. By (20.7), what is the smallest standard error an unbiased estimator of the annual mean can have?

ANSWER

For $T = 246$, $\mathrm{Var} \geq \sigma^2/246$, so the annual mean's standard error is at least $A\sigma/\sqrt{246}$, 18.96 points, which the sample mean attains.

7 THE DELTA METHOD

- [ ] apply the delta method to the annualized volatility, the Kelly fraction and the Sharpe ratio of independent returns

The annual volatility, the Kelly fraction and the Sharpe ratio are smooth functions of $\bar{R}$ and s^2. Their standard errors follow from a first-order expansion.

THEOREM **Delta method**

Let $\hat{\boldsymbol{\theta}}$ be approximately normal with mean $\boldsymbol{\theta}$ and covariance matrix $\operatorname{Cov}(\hat{\boldsymbol{\theta}})$, and let h be differentiable at $\boldsymbol{\theta}$ with $\nabla h \neq \mathbf{0}$. Then $h(\hat{\boldsymbol{\theta}})$ is approximately normal with mean $h(\boldsymbol{\theta})$ and

$$\operatorname{Var} h(\hat{\boldsymbol{\theta}}) \approx \nabla h^{\top} \operatorname{Cov}(\hat{\boldsymbol{\theta}})\, \nabla h. \tag{20.8}$$

Why. By (7.1), $h(\hat{\boldsymbol{\theta}}) \approx h(\boldsymbol{\theta}) + \nabla h^{\top}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})$, and (16.4) gives the variance of this linear function, as (10.4) gave a portfolio's (Wasserman 2004).

Up to a term of order $1/T$, s^2 averages the iid terms $(R_t - \mu)^2$, each with variance $(\operatorname{Kurt} - 1)\sigma^4$ by (14.5). So $\operatorname{Var}(s^2) \approx (\operatorname{Kurt} - 1)\sigma^4/T$ by (17.1), and $h(v) = \sqrt{v}$, with derivative $1/(2\sigma)$ at σ^2, gives by (20.8)

$$\operatorname{SE}(s) \approx \sigma \sqrt{\frac{\operatorname{Kurt} - 1}{4T}},$$

which needs a finite fourth moment. For normal days $\operatorname{Kurt} = 3$, and this is Example 7's $\sigma/\sqrt{2T}$.

EXAMPLE 8

How precisely TAIEX's volatility is known

Give standard errors and 95% intervals for TR23's annual volatility, LM19's 18.96%, and for the standard deviation of the 23 annual log returns, LM15's 22.207%.

SOLUTION

With LM14's kurtosis 8.27, the volatility's standard error is $18.96 \times \sqrt{7.27/(4 \times 5{,}849)} = 0.33$ points, so the interval is 18.3% to 19.6%. For normal days it is $18.96/\sqrt{2 \times 5{,}849} = 0.175$ points.

For 23 normal years, $22.207/\sqrt{46} = 3.27$ points gives 15.8% to 28.6%. LM17's daily-based 18.563% (standard error 0.34, from the kurtosis 8.55 of its 5,664 log returns) lies 3.6 points, or 1.1 standard errors, below. So LM19's gap is within the sampling error of 23 values.

Under a normal model of the years, $23s^2/\sigma^2 = 22s_{\mathrm{u}}^2/\sigma^2 \sim \chi^2_{22}$ (Blitzstein and Hwang 2019, Example 10.4.3). Its 2.5% and 97.5% quantiles are 10.98 and 36.78 (`stats.chi2.ppf`), so $22.207\% \times \sqrt{23/36.78}$ to $22.207\% \times \sqrt{23/10.98}$ gives 17.6% to 32.1%.

Unlike the mean's, the volatility's standard error does not survive clustering: large squared returns cluster (LM19), so the 0.33 is probably too small (LM22). By regime, s is 1.257% (0.026) before 1 June 2015 and 1.152% (0.035) after: a stability check, not a test (LM21).

For the Kelly fraction and the Sharpe ratio, $\hat{\boldsymbol{\theta}} = (\bar{R}, s^2)$, with $\operatorname{Cov}(\bar{R}, s^2) \approx \operatorname{Skew}\, \sigma^3/T$ by (14.5). The gradients of μ/σ^2 and μ/σ in (μ, σ^2) are $(1/\sigma^2, -\mu/\sigma^4)$ and $(1/\sigma, -\mu/(2\sigma^3))$, and (20.8) gives the next result.

RESULT **Standard errors of the Kelly fraction and the Sharpe ratio**

Let the days be iid with $\mathbb{E}R^4 < \infty$ and $\mathrm{SR} = \mu/\sigma$. For large T,

$$\begin{aligned} \sigma^2\, T\,\mathrm{Var}(\hat{f}) &\approx 1 - 2\,\mathrm{SR}\,\mathrm{Skew} \\ &\quad + \mathrm{SR}^2(\mathrm{Kurt} - 1), \\ T\,\mathrm{Var}\left(\widehat{\mathrm{SR}}\right) &\approx 1 - \mathrm{SR}\,\mathrm{Skew} \\ &\quad + \tfrac{1}{4}\mathrm{SR}^2(\mathrm{Kurt} - 1). \end{aligned} \tag{20.9}$$

EXAMPLE 9

How far the Kelly fraction can be trusted

Give standard errors and 95% intervals for TR23's $\hat{f} = \bar{R}/s^2$ and its Sharpe ratio. Then redo LM4's band for 0050 ($\hat{\mu} = 0.0871\%$, $s = 1.2385\%$, $T = 2{,}902$).

SOLUTION

TR23's $\hat{f} = 0.0625\%/1.2084\%^2 = 4.28$. By (20.9) with $\widehat{\mathrm{SR}} = 0.0517$, skewness −0.34 and kurtosis 8.27, its standard error is 1.11 and the interval 2.10 to 6.46 (Exhibit 8). The mean's error alone, $1/(s\sqrt{T}) = 1.08$, gives most of it.

LM19's daily Sharpe ratio, 0.0517, has standard error 0.0132. Times $\sqrt{A}$, the annual 0.81 has 0.21 and the interval 0.40 to 1.22.

For 0050, $\hat{\mu}/s^2 = 5.68$. The mean's error alone gives 1.50 and the band 2.74 to 8.62: LM4's quadratic 2.7 to 8.6 was the delta method with the mean's error only. The full (20.9) gives 1.53 and 2.68 to 8.68. LM4's exact band, 2.7 to 7.6, is skewed, which no delta interval can show (LM22).

Exhibit 8: Delta-method standard errors under independent days

Statistic, sample	Estimate	SE, mean only	SE, normal	SE, plug-in	95% interval, plug-in
Annual volatility, %, TR23	18.96	—	0.175	0.334	18.3 to 19.6
Kelly fraction $\hat{f}$, TR23	4.28	1.082	1.085	1.111	2.10 to 6.46
Kelly fraction $\hat{f}$, 0050	5.68	1.499	1.506	1.532	2.68 to 8.68
Sharpe ratio a day, TR23	0.0517	—	0.0131	0.0132	0.0258 to 0.0776
Sharpe ratio a year, TR23	0.81	—	0.205	0.207	0.40 to 1.22

Note: iid days, daily simple returns, $r_f = 0$. "Mean only" keeps the first term of (20.9), "normal" uses skewness 0 and kurtosis 3, and "plug-in" the sample's: −0.34 and 8.27 for TR23, 0.04 and 11.12 for 0050. These four moments are printed without standard errors, which would need the sixth and eighth moments. The volatility, 0.0517 and 0.81 are LM19's estimates.

0050 is LM4's sample of 2,902 total returns on the days both 0050 and 00631L traded (two span a fund's split halt), 3 November 2014 to 8 October 2026. Sources: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026; `TaiwanStockPrice` (0050, 00631L) and `TaiwanStockDividendResult` (0050), retrieved 10 October 2026.

For a statistic with no formula, such as LM11's eigenvalue share, LM22's bootstrap replaces the delta method. The Sharpe ratio of dependent returns is LM33's, since $\sqrt{A}$ annualizes it exactly only for log returns on uncorrelated days (LM19; Lo 2002). Sizing with an uncertain fraction is LM45's.

> **PITFALL** **A plug-in standard error needs the moments it plugs in**
>
> Under the Student-t fit, $\nu = 3.065 < 4$ and $\mathbb{E}R^4 = \infty$: (20.9)'s condition fails, and the sample kurtosis estimates an infinite quantity. The 0.33 and the 1.11 rest on it and are probably too small. Exhibit 3's volatility errors and the regimes' 0.026 and 0.035 rest on the sample kurtosis too. The skewness and the kurtosis get no standard error at all.

導讀 **delta 法：函數的標準誤**

年化波動率（annualized volatility）、凱利比例（Kelly fraction）與夏普比率（Sharpe ratio）都是樣本平均數（sample mean）與樣本變異數（sample variance）的平滑函數（smooth function），它們的標準誤（standard error）可以用 delta 法（delta method）取得：把函數在真值附近做一階展開（first-order expansion），變成線性組合（linear combination），再用共變異數矩陣（covariance matrix）算變異數（variance）。加權股價報酬指數的年化波動率 18.96%，標準誤約 0.33 個百分點；凱利比例 4.28，標準誤 1.11，95% 信賴區間（confidence interval）2.10 到 6.46；年化夏普比率 0.81，標準誤 0.21。這些都假設日報酬獨立同分布（iid），而且四階動差（fourth moment）有限；在自由度（degrees of freedom）3.065 的 t 分布（Student-t distribution）下四階動差無限大，所以這些數字可能偏小。23 個年報酬的標準差（standard deviation）22.207%，標準誤高達 3.27 個百分點，足以解釋它與日資料推算的 18.563% 之間的差距。

KNOWLEDGE CHECK 6

Why is the standard error of an annual Sharpe ratio about $1/\sqrt{\tau}$, and what is it for TR23?

ANSWER

For a small daily SR, (20.9) gives $T\,\mathrm{Var}(\widehat{\mathrm{SR}}) \approx 1$. Times $\sqrt{A}$, the standard error is $\sqrt{A/T} = 1/\sqrt{\tau}$: 0.205 for TR23's 23.7645 years.

8 THE HILL ESTIMATOR OF A TAIL INDEX

- ☐ estimate a tail index with the Hill estimator and its standard error, and explain how the threshold trades bias against variance

LM15 read TAIEX's tail from chords between loss levels, 2.09 to 5.03, and its Student-t fit implies a tail index of $\nu = 3.065$. A tail index can also be estimated from the largest losses alone.

Let $X = -R$ be a day's loss and $x_0 > 0$ a loss level, the **threshold**. Model the losses beyond x_0 as an exact power tail, $\Pr(X > x \mid X \geq x_0) = (x/x_0)^{-\alpha}$ for $x \geq x_0$. This is a **Pareto distribution**, with density $\alpha x_0^{\alpha} x^{-\alpha-1}$ on $x \geq x_0$. Its support does not depend on α, so (20.6) applies. For k iid losses $x_1, \ldots, x_k \geq x_0$,

$$\ell(\alpha) = k \ln \alpha + k\alpha \ln x_0 - (\alpha + 1) \sum_{i=1}^{k} \ln x_i,$$

$$\ell'(\alpha) = \frac{k}{\alpha} - \sum_{i=1}^{k} \ln \frac{x_i}{x_0}.$$

The root of ℓ' is a maximum, since $\ell'' = -k/\alpha^2 < 0$. One loss carries the information $1/\alpha^2$.

RESULT **Hill estimator**

Let the losses at or beyond x_0 be iid with an exact power tail of index α, and let k of them be observed. The MLE of α and its standard error are

$$\hat{\alpha} = \frac{k}{\sum_{i=1}^{k} \ln(x_i / x_0)}, \qquad \mathrm{SE}(\hat{\alpha}) \approx \frac{\hat{\alpha}}{\sqrt{k}}. \tag{20.10}$$

Hill (1975) based tail inference on the conditional likelihood given the largest order statistics; here x_0 is a fixed loss level, as in LM15's counts.

EXAMPLE 10

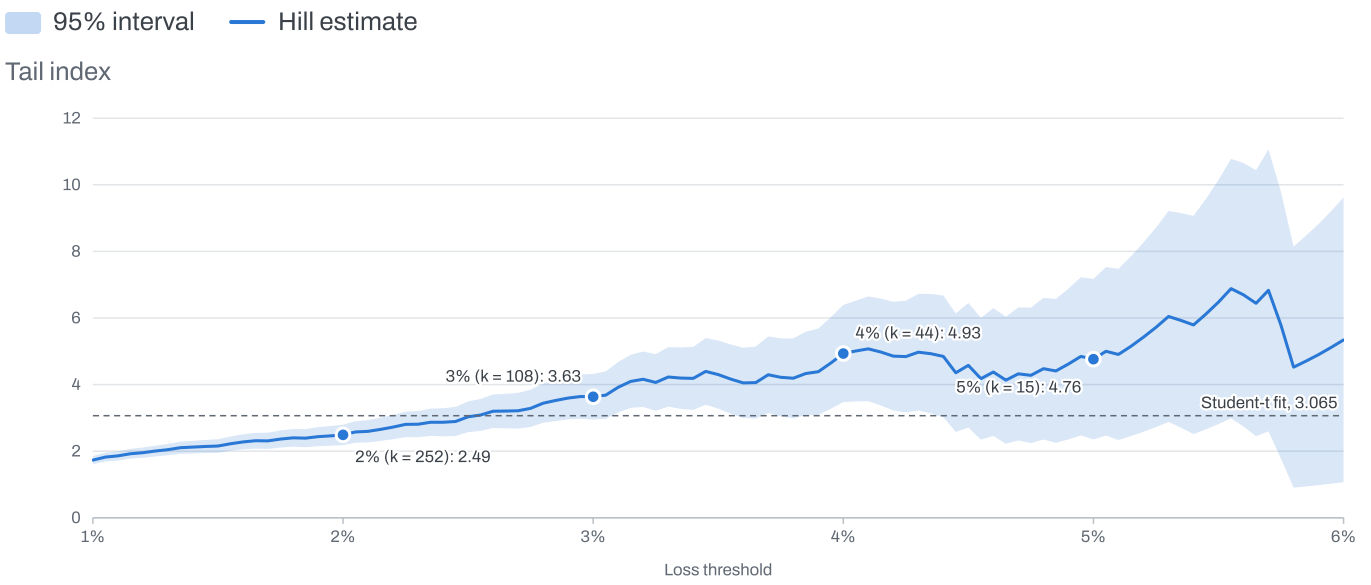
TAIEX's tail index

Estimate the tail index of TR23's daily losses at $x_0 = 3\%$, and then at 2%, 4% and 5%.

SOLUTION

At 3% the guard `R <= -0.03 + 1e-12` finds $k = 108$ losses, LM13's count. By (20.10), $\hat{\alpha} = 3.63$ with standard error $3.63/\sqrt{108} = 0.35$, so the interval is 2.9 to 4.3. At 2%, 4% and 5% the estimates are 2.49 (0.16), 4.93 (0.74) and 4.76 (1.23), from k = 252, 44 and 15 (Exhibit 9). The estimate rises from 2% to 4%, as LM15's chords did, then wanders once k falls below 45. The Student-t's 3.065 sits near the low thresholds.

Exhibit 9: Hill estimates of TAIEX's tail index by threshold



Note: TR23's daily losses at or beyond x_0, for x_0 from 1.0% to 6.0% in steps of 0.05 points, counted with the guard; band $\hat{\alpha} \pm 1.96\,\hat{\alpha}/\sqrt{k}$. Marked: 2%, 3%, 4% and 5%, with k = 252, 108, 44 and 15. The dashed line is the Student-t fit's 3.065. The 7% limit before June 2015 and the 10% limit since cap the far tail. In-sample. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026.

PITFALL **The threshold trades bias against variance**

A low threshold brings many losses and a small standard error, but losses near 2% belong to the body of the distribution, not to a power tail. A high one fits the tail model better but leaves few losses. Big losses also come in runs (LM19), so the iid $\hat{\alpha}/\sqrt{k}$ is likely too small. Value at risk from a tail model is LM32's; extreme value theory beyond Hill's estimator is not in this book.

導讀 **希爾估計量：尾部指數**

尾部指數（tail index）描述極端損失出現的頻率隨幅度遞減的速度，數值越小尾巴越厚。若門檻（threshold）x_0 以上的損失服從帕累托分布（Pareto distribution），最大概度估計（maximum likelihood estimation）就是希爾估計量（Hill estimator）$k/\sum \ln(x_i/x_0)$，標準誤（standard error）約為 $\hat{\alpha}/\sqrt{k}$。加權股價報酬指數在 3% 門檻有 108 個損失日，估計值（estimate）3.63，標準誤 0.35；門檻 2% 時是 2.49，4% 時升到 4.93。門檻低則樣本多、標準誤小，卻混入不屬於尾部（tails）的損失而產生偏誤（bias）；門檻高則偏誤小、變異數（variance）大。大跌日又常成群出現，所以獨立（independent）假設下的標準誤可能偏小。

KNOWLEDGE CHECK 7

At $x_0 = 5\%$, $k = 15$ and the standard error is 1.23. How wide is the 95% interval?

ANSWER

The interval's width is $2 \times 1.96 \times 1.23 = 4.8$, almost five units of tail index.

9 ESTIMATION IN PYTHON

- [ ] compute estimates, standard errors and intervals in Python with explicit divisors and seeded simulations

Every estimate here takes a few library calls; the divisor, the tolerance and the seed must be explicit.

PYTHON

```python
import numpy as np
import polars as pl
from scipy import optimize, stats

tr = pl.read_csv("data/taiex_tr.csv")["price"].to_numpy()
R = tr[1:] / tr[:-1] - 1          # T = 5,849 simple returns
T = len(R)
s, s_u = R.std(ddof=0), R.std(ddof=1)   # 1.208364%, 1.208468%
se_mean = s / np.sqrt(T)                # 0.01580% a day
q = stats.t.ppf(0.975, T - 1)           # 1.96037

x = 100 * R                       # percent, for the optimizer


def nll(p):                       # minus the Student-t log-likelihood
    return -stats.t.logpdf(x, *p).sum()


fit = optimize.minimize(nll, stats.t.fit(x), method="Nelder-Mead",
                        options={"xatol": 1e-12, "fatol": 1e-12})


def hessian(f, p, h):             # central differences
    k = len(p)
    H = np.zeros((k, k))
    for i in range(k):
        for j in range(k):
            a, b = np.eye(k)[i] * h[i], np.eye(k)[j] * h[j]
            H[i, j] = (f(p + a + b) - f(p + a - b) - f(p - a + b)
                       + f(p - a - b)) / (4 * h[i] * h[j])
    return H


for h in ([1e-4, 1e-5, 1e-5], [1e-3, 1e-4, 1e-4]):   # two step sizes
    cov = np.linalg.inv(hessian(nll, fit.x, h))
    print(np.sqrt(np.diag(cov)))  # 0.144, 0.0124, 0.0138 both times

rng = np.random.default_rng([1, 20261011])   # experiment 1, once
sims = R.mean() + s * rng.standard_normal((10_000, 246))
vol = np.sqrt(T / 23.7645) * sims.std(axis=1, ddof=0)
print(np.quantile(vol, [0.025, 0.975], method="inverted_cdf"))
```

NumPy's `std` divides by T by default and Polars' by $T-1$, so pass `ddof` every time. SciPy's Nelder–Mead stops at a tolerance of 10^{-4} unless told otherwise. The last lines repeat the normal block of Exhibit 2: 17.3% to 20.6%.

IN PYTHON **Companion lab**

The Polars-first notebook `LM20_lab.ipynb` rebuilds the module's exhibits and computational examples and solves the computational practice problems. It ends with the Kelly fraction and its delta-method interval in each limit regime, a stability check.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 8

`pl.col("R").std()` returns 1.208468% for TR23, NumPy's `R.std()` 1.208364%. Which is s?

ANSWER

NumPy's: it divides by T, while Polars divides by $T-1$ by default and returns s_{u}.

SUMMARY

- Mean squared error is variance plus squared bias, (20.1); a biased estimator can win.
- A mean's standard error is $\sigma/\sqrt{T}$, (20.2); annualized, the years of data decide it. Divisor T biases s^2 by $(T-1)/T$, (20.3).
- The interval $\bar{R} \pm 1.96\, s/\sqrt{T}$ needs a large T; the t_{T-1} interval is exact for normal days, (20.4).
- Maximum likelihood gives $\bar{R}$ and s for the normal model, (20.5), and fits a Student-t numerically.
- The observed information gives standard errors, (20.6); no unbiased estimator beats the Cramér–Rao bound, (20.7).
- The delta method, (20.8) and (20.9), gives the volatility, the Kelly fraction and the Sharpe ratio standard errors under iid days and a finite fourth moment.
- The Hill estimator, (20.10), estimates a tail index; its threshold trades bias against variance.

PRACTICE PROBLEMS

1. Ten years of independent daily returns have an annual volatility of 20% ($A = 246.12$). The annual mean's standard error, from daily or from five-minute returns (54 a session), is *closest* to:

   A. 6.32 points either way

   B. 6.32 points from daily returns, 0.86 points from five-minute returns

   C. 0.40 points either way

2. Twenty-three annual log returns have standard deviation s with divisor 23. By what percentage of s does s_{u} exceed s?

   A. 4.5%

   B. 0, because s is unbiased

   C. 2.2%

3. "7.8% to 23.0% is a 95% confidence interval for TR23's annual mean $A\mu$." This means:

   A. the mean $A\mu$ lies in it with probability 0.95.

   B. intervals built this way cover $A\mu$ in 95% of samples, if the days are iid.

   C. 95% of future annual returns will fall in it.

4. Under a normal model, the maximum-likelihood estimate of the standard deviation of the 23 annual log returns of 2003–2025 is *closest* to:

   A. 22.71%

   B. 18.56%

   C. 22.21%

5. For iid normal days, the standard error of TR23's annual volatility $\sqrt{A}\,s = 18.96\%$, with $T = 5{,}849$, is *closest* to (two decimals):

   A. 0.33 points

   B. 0.18 points

   C. 0.25 points

6. Let (X, Y) be a standard normal pair with correlation ρ. (a) Use (16.8) to show that $\mathbb{E}[X^3Y] = 3\rho$ and $\mathbb{E}[X^2Y^2] = 1 + 2\rho^2$. (b) The delta method gives $\hat\rho - \rho \approx \frac{1}{T}\sum_t L_t$ with $L_t = X_tY_t - \frac{\rho}{2}(X_t^2 + Y_t^2)$. Show that $\operatorname{Var}\hat\rho \approx (1-\rho^2)^2/T$. (c) *(Python)* For LM16's Semiconductor and FinancialInsurance (U28 price indices, no dividends, $T = 4{,}382$, divisor T), $\hat\rho = 0.5749$. Give the normal-theory standard error and the plug-in $\mathrm{sd}(L_t)/\sqrt{T}$ of standardized returns (four decimals), with intervals (three). Why do they differ?

7. *(Python)* Fit the Student-t to TR23's 3,077 returns before 1 June 2015 and its 2,772 after, and give ν with its observed-information standard error (three decimals). Give the difference in ν and its standard error, the fits taken as independent. Is this a test?

8. In each of TR23's 286 calendar months, count the days with $|R| \geq 3\%$ (guard `1e-12`), zero counts included. (a) Derive the Poisson MLE $\hat\lambda$ and its information-based standard error. (b) Compute both (five decimals). (c) Compare with the counts' standard deviation (divisor 286) over $\sqrt{286}$.

9. TR23 has 3,205 up days ($R > 10^{-12}$) among its 5,849. Model the days as iid trials with up probability θ. Derive $\hat\theta$ and $\mathcal{I}(\theta)$, and give $\hat\theta$ and its standard error (four decimals) and the 95% interval (three).

10. Compute the Hill estimates of TR23's tail index at $x_0 = 2\%$ and 4%, with standard errors (two decimals) and 95% intervals (one decimal). Why do the two intervals not overlap?

11. *(Python)* Experiment 3: with `default_rng([3, 20261011])`, draw `standard_normal((20000, 5))`. Give the coverage of $\bar{X} \pm 1.96\, s/\sqrt{5}$ (divisor 5) and of the t_4 interval with s_{u}, with Monte Carlo standard errors, and the first one's exact coverage. Give coverages in percent to one decimal, and standard errors in points to two. Is the gap between the first interval's simulated and exact coverage chance?

12. Under iid normal daily returns with $A = 246.12$, an annual Sharpe ratio of 0.5 has a standard error of 0.1 after *closest* to:

    A. 100 years

    B. 10 years

    C. 384 years

SOLUTIONS

1. **A** is correct. By (20.2), the annual mean's standard error is $20\%/\sqrt{10} = 6.32$ points, whatever the sampling interval. B divides by $\sqrt{54}$, as if five-minute returns told more about the mean. C is $20\%/\sqrt{2{,}461}$: it treats each of the 2,461 days as a year.

2. **C** is correct: $\sqrt{23/22} - 1 = 2.2\%$. A is the variance's factor, $23/22 - 1 = 4.5\%$. B ignores the divisors: $s_{\mathrm{u}}/s = \sqrt{23/22}$ in every sample.

3. **B** is correct: the 95% is the coverage of the procedure, with $A\mu$ fixed. A treats $A\mu$ as random, a posterior statement (LM24). C confuses the mean with single years, which spread about 22 points around it (LM15).

4. **C** is correct: by (20.5), the MLE is s with divisor 23, 22.21%. A is s_{u}, 22.71%, which is not the MLE. B is LM17's daily-based 18.56%, another estimator on other data.

5. **B** is correct: $18.96/\sqrt{2 \times 5{,}849} = 0.18$ points. A uses the sample kurtosis 8.27, not the stated normal model. C forgets the 2: $18.96/\sqrt{5{,}849} = 0.25$.

6. (a) By (16.8) with unit variances, $Y = \rho X + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, 1-\rho^2)$ independent of X. So $\mathbb{E}[X^3 Y] = \rho \mathbb{E}X^4 = 3\rho$ and $\mathbb{E}[X^2 Y^2] = \rho^2 \mathbb{E}X^4 + \mathbb{E}X^2\, \mathbb{E}\varepsilon^2 = 3\rho^2 + 1 - \rho^2 = 1 + 2\rho^2$, by (14.6). (b) The mean of L_t is $\rho - \rho = 0$. With (a) and $\mathbb{E}[(X^2+Y^2)^2] = 8 + 4\rho^2$,

$$\begin{aligned} \mathbb{E}L_t^2 &= 1 + 2\rho^2 - 6\rho^2 + \tfrac{\rho^2}{4}(8 + 4\rho^2) \\ &= (1-\rho^2)^2, \end{aligned}$$

   and (17.1) gives $\operatorname{Var}\hat{\rho} \approx (1-\rho^2)^2/T$. (c) The normal-theory standard error is $(1 - 0.5749^2)/\sqrt{4{,}382} = 0.0101$, so the interval is 0.555 to 0.595. The plug-in one is 0.0153, with interval 0.545 to 0.605, 1.51

times wider. Fat tails and clustered volatility make $\mathbb{E}[X^2Y^2]$ larger than the normal pair's. The plug-in error also needs finite fourth moments, which daily returns may lack (LM19); it is nominal.

7. The regimes give $\nu = 2.990$ (0.195) and 3.237 (0.223). The difference, 0.247, has standard error $\sqrt{0.195^2 + 0.223^2} = 0.296$, so it is about 0.8 standard errors. It is not yet a test: that needs a null hypothesis, a reference distribution and a count of the tests run (LM21).

8. (a) For counts $j_1, \dots, j_{286}$, $\ell(\lambda) = \sum_i \left(j_i \ln \lambda - \lambda - \ln j_i!\right)$, so $\ell'(\lambda) = \sum_i j_i/\lambda - 286 = 0$ gives the mean count. Per month $\mathcal{I}(\lambda) = 1/\lambda$, so the standard error is $\sqrt{\hat\lambda/286}$. (b) There are 175 such days, so $\hat\lambda = 175/286 = 0.61189$ and the standard error is 0.04625. (c) The counts' own standard deviation gives 0.0803, $\sqrt{3.0167} = 1.737$ times larger: their variance is 3.0167 times their mean (LM15's 3.02). Large moves cluster, the Poisson model is wrong, and its information understates the error.

9. With j up days, $\ell(\theta) = j \ln\theta + (T - j)\ln(1-\theta)$, so $\hat\theta = j/T = 3{,}205/5{,}849 = 0.5480$ and $\mathcal{I}(\theta) = 1/\big(\theta(1-\theta)\big)$. The standard error is $\sqrt{\hat\theta(1-\hat\theta)/T} = 0.0065$, so the interval is 0.535 to 0.561. LM17's 9,604 was a model's trial count; this is an interval for a data frequency, under iid days. LM21 tests $\theta = \frac{1}{2}$.

10. At 2%, $k = 252$ and $\hat\alpha = 2.49$ (0.16), interval 2.2 to 2.8. At 4%, $k = 44$ and $\hat\alpha = 4.93$ (0.74), interval 3.5 to 6.4. If both thresholds lay in one exact power tail, the intervals would estimate the same α. Losses near 2% belong to the body, which biases the low estimate down; the high one has the larger variance. Clustered losses make both iid standard errors too small (Section 8), so the intervals are also too narrow.

11. The 1.96 interval covers in 84.3% of samples (Monte Carlo standard error 0.26) and the t_4 interval in 94.8% (0.16). The exact coverage of the first is $2F_{t_4}(1.96\sqrt{4/5}) - 1 = 84.6\%$. Before rounding, the gap is 0.23 points (84.55% against 84.32%), about one Monte Carlo standard error: chance. The t_4 interval's 94.77% lies about one and a half standard errors from 95%, also chance.

12. **A** is correct. With $\mathrm{SR} = 0.5/\sqrt{246.12}$ a day, (20.9) for normal days gives an annual standard error of $\sqrt{(1 + \mathrm{SR}^2/2)/\tau}$, which is 0.1 at $\tau = 100.05$ years. B forgets the square root, $1/0.1$. C takes 0.1 as the half-width of the interval, $1.96^2/0.1^2$.

GLOSSARY

Term	中文	Meaning
Estimator	估計量	A function of the sample that estimates a parameter; its value on the data is the estimate.
Sampling distribution	抽樣分布	The distribution of an estimator across samples.
Bias	偏誤	$\mathbb{E}\hat{\theta} - \theta$.
Unbiased estimator	不偏估計量	An estimator with bias 0.
Mean squared error	均方誤差	$\mathbb{E}[(\hat{\theta} - \theta)^2]$, variance plus squared bias, (20.1).
Consistent estimator	一致估計量	An estimator that converges in probability to the parameter.
Standard error	標準誤	The standard deviation of an estimator, (20.2) for a mean.
Confidence interval	信賴區間	An interval rule that covers the parameter with a stated probability across samples, (20.4).
Likelihood function	概度函數	$\mathcal{L}(\theta)$, the density of the observed sample as a function of the parameters.
Maximum likelihood estimation	最大概度估計	Choosing the parameters that maximize the likelihood, (20.5).
Fisher information	費雪訊息	$\mathcal{I}(\theta)$, the expected curvature of one observation's log-likelihood.
Observed information matrix	觀察訊息矩陣	$-H$, the negative Hessian of ℓ at the estimate; its inverse estimates the covariance, (20.6).
Asymptotic normality	漸近常態性	The approximately normal law of an estimator in large samples, (20.6).
Profile likelihood	剖面概度	The likelihood of one parameter, maximized over the others at each of its values; its logarithm is the profile log-likelihood.
Cramér–Rao lower bound	克萊姆-饒歐下界	$1/(T\mathcal{I})$, below which no unbiased estimator's variance falls, (20.7).
Efficiency	效率	Precision of an estimator relative to the Cramér–Rao bound or to another estimator.
Delta method	delta 法	Standard errors of smooth functions by a first-order expansion, (20.8).

Term	中文	Meaning
Pareto distribution	帕累托分布	The exact power tail $\Pr(X > x \mid X \geq x_0) = (x/x_0)^{-\alpha}$.
Hill estimator	希爾估計量	The Pareto MLE of a tail index from the losses beyond a threshold, (20.10).

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